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A Cambrian pole for the Moroccan Coastal Meseta

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SUMMARY

Middle Cambrian volcanics of Sidi-Saïd-Maâchou, East of El Jadida, western Morocco (158 cores, 23 sites), sampled in two localities North of Sidi-Saïd-Maâchou, were subjected to progressive alternating field and thermal demagnetization, thermomagnetic cycles, isothermal remanent magnetization acquisition experiments as well as microscopic observations.

The magnetic mineralogy of these volcanic rocks is dominated by magnetite and titanomagnetites. The palaeomagnetic analysis has revealed two characteristic components of magnetization: the first (direction $D = 300°$, $I = -31°$, $\alpha_{95} = 6°$, pole position at 14°N, 231°E, $\delta p = 4°$, $\delta m = 7°$, palaeolatitude = 16.7°S) is carried by magnetite and yields a positive fold test. It is interpreted, using the results of a baked contact test, as the original magnetization of Middle Cambrian age (520 Ma); the second (direction $D = 121°$, $I = 20°$, $\alpha_{95} = 6°$, pole position at 18°S, 55°E, $\delta p = 3°$, $\delta m = 6°$) is nearly anti-parallel to the first magnetization but the fold test is not conclusive, and it is interpreted as an overprint that occurred a short time after this volcanism. This study provides the first detailed palaeomagnetic result from Cambrian rocks in the Moroccan Meseta.

Key words: Cambrian, Morocco, Nova Scotia, palaeogeography, palaeolatitude, palaeomagnetism.

INTRODUCTION

The Palaeozoic palaeogeography of Gondwana is presently a subject of considerable debate. A palaeogeographical map can be constructed from such distinct disciplines of the earth sciences as palaeomagnetism, palaeontology, palaeoclimatology, etc., each contributing to the knowledge of palaeolatitude. Each of these approaches, however, has its drawbacks and uncertainties, which are discussed in several papers (e.g. Van der Voo 1988; Paris & Robardet 1990). The comparison between the reliable data from all these approaches is the best way to constrain the different models.

A large palaeomagnetic study was undertaken on Palaeozoic rocks of Morocco (Khattach 1989; Khattach, Perroud & Robardet 1989) and this paper gives the results of a palaeomagnetic study on Middle Cambrian volcanic rocks from the Coastal Meseta in western Morocco. This region was chosen because it is very weakly deformed and therefore suitable for palaeomagnetic analysis. This study attempts to improve the understanding of the palaeogeography during Cambrian times.

GEOLOGICAL CONTEXT

The volcanics of Sidi-Saïd-Maâchou, located about 50 km east of El Jadida (Fig. 1), were first described by Gigout (1956). These rocks crop out in the valley of Oum-er-Rbia River, north of Sidi-Saïd-Maâchou (especially at the base of Sidi-Moul-Nker) and in some tributary valleys (particularly near K. Sbaâ). The volcanic rocks are included in a shaly–silty succession with green shales overlain by purple shales with trilobite-bearing pink limestones levels. Purple shales and siltstones are in close relation with lava flows and sills: they appear with the first lava flows and their last occurrence slightly precedes the end of the volcanic activity. The whole succession is cut by vertical dykes. Gigout (1956) described a section with dykes, sills and lava flows at the base of Sidi-Moul-Nker.

(1) The dykes consist of grey-blue fine-grained rocks with abundant magnetite, biotite, calcite or serpentine aggregates and minute but well-preserved plagioclases.

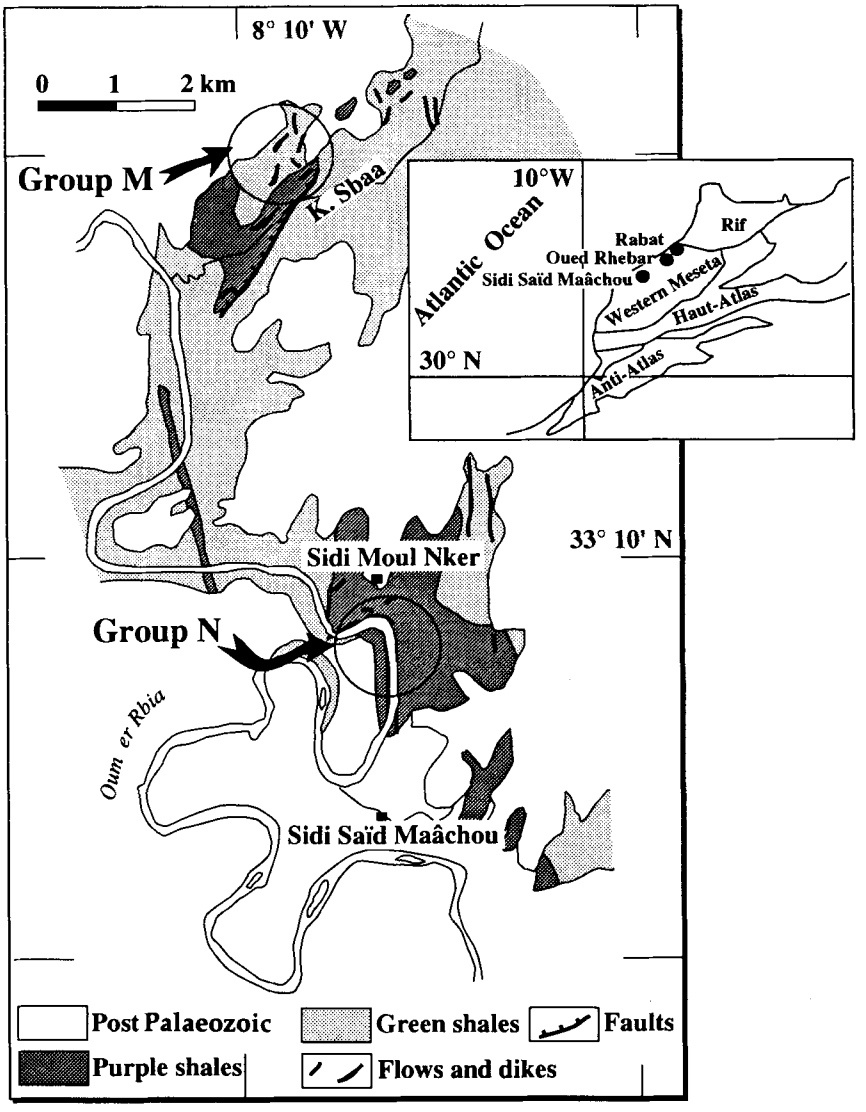


Figure 1. Simplifiéd geological map of the Sidi-Saïd-Maâchou area (after Gigout 1956), and localization of the sampling sites (groups M and N), together with the main structural domains of Morocco.

(2) The sills closely resemble the preceding rocks with numerous magnetite grains.
(3) The lava flows consist of fine-grained rocks with small scattered feldspathic and calcitic crystals. Under the microscope these rocks show an ophiolitic structure with vacuoles filled up by spherolitic fibrous minerals and calcitic crystal in the central part; opaque mineral grains are oligiste and ilmenite.

Some outcrops show irregularities in the stratification of the lava flows, with small linear flexures filled by limestones. Within the volcanic rocks, small vacuoles correspond to gas escapes. These characteristics indicate the submarine nature of the volcanic manifestations. The fossiliferous limestones (with trilobites such as Paradoxides) clearly indicate a Middle Cambrian age (Acadian). It is difficult to estimate accurately the size of the volcanic apparatus, due to the overlying post-Palaeozoic formations. However Gigout (1956), considering the volcanic components included in the Cambrian sediments, proposed a diametrical extension of about 10 km and a total thickness (including the sedimentary rocks) of about 1500 m.

It must be noted that feldspars are neither weathered nor sericitized and that microlites are well preserved. Calcite and chlorite that fill up the vacuoles of the rock result from destabilization processes contemporaneous with the volcanic activity. Thin sections of the rocks show no trace of structuration or metamorphism.

The Cambrian formations were moderately folded and faulted (N–S faults) during Hercynian (Variscan) times and the average deformation does not exceed 12 per cent (Bernardin *et al.* 1988).

SAMPLING

Sampling was at two groups of sites: M (nine sites: MQ to MY) and N (14 sites: NQ to NZ) (Fig. 1). A total of 158 cores were drilled using a portable gas-powered drill and

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oriented with both magnetic and solar compasses. Sites were chosen in order to allow the determination of bedding with sediments. In addition, sites having different structural attitudes were sampled so that a palaeomagnetic fold test could be performed. The N series, located along Oum-er-Rbia River to the north of S.S. Maâchou village, has an eastern dip which is opposite to the western dip of the M series located near K. Sbaâ. The value of the dip fluctuates between 10° and 60°.

The host rock of a dyke (site NT) was also sampled for a second palaeomagnetic test (baked contact test) to help define the age of magnetizations.

LABORATORY METHODS

Standard palaeomagnetic specimens (24 mm high) were prepared from the oriented core samples, and their remanent magnetization was measured with a Schonstedt DSM-1 Spinner magnetometer. Specimens were demagnetized by stepwise alternating-field (AF) and thermal methods, using a Schonstedt GSD-1 AF demagnetizer and a Schonstedt TSD-1 furnace, and results for each specimen were plotted on orthogonal vector diagrams (Zijderveld 1967) as well as on equal-area projections. Demagnetization trajectories were interpreted by fitting straight lines through the data points. Great circles were used for a few samples (site NT) when planar demagnetization trajectories were identified.

The magnetic carriers in these volcanic rocks were investigated using a number of different rock magnetic techniques, as well as microscopic observations. Isothermal remanent magnetization (IRM) acquisition experiments were performed on unheated core specimens at room temperature (20 °C), in applied fields of up to 1.2 tesla (T). This IRM was demagnetized by a stepwise thermal method. Thermomagnetic experiments were also conducted with an automated Brucker Curie balance.

PALAEOMAGNETIC RESULTS

A short-period viscosity test (Thellier & Thellier 1959) was conducted before any demagnetization. It entailed storing the specimens oriented in the laboratory field for a short duration (typically 10 days), then measuring natural remanent magnetization (NRM), storing again in reverse orientation for the same duration and measuring again; this allows the determination of the true remanent magnetization intensity, direction (free of short-term viscosity), and the viscosity index (ratio of intensities of viscous and remanent magnetizations), which is a helpful indicator in the interpretation of the soft components of magnetization. A total of 130 specimens were analysed in that way and the results are displayed in Figs 2 and 3, in the form of a density contour plot for the directions, and bar diagrams for the NRM intensities and viscosity indexes.

NRM intensities (Fig. 2a) show a rather large dispersion, between 10^{-4} and $10\,A\,m^{-1}$ with one mode at $3\times 10^{-2}\,A\,m^{-1}$. The values less than $10^{-3}\,A\,m^{-1}$ correspond to the sediment host rock (site NT). Viscosity indices (Fig. 2b) are distributed between 0 and 60 per cent, with a large majority (84 per cent) below 20 per cent.

NRM directions plotted (Fig. 3) in geographic coordinates

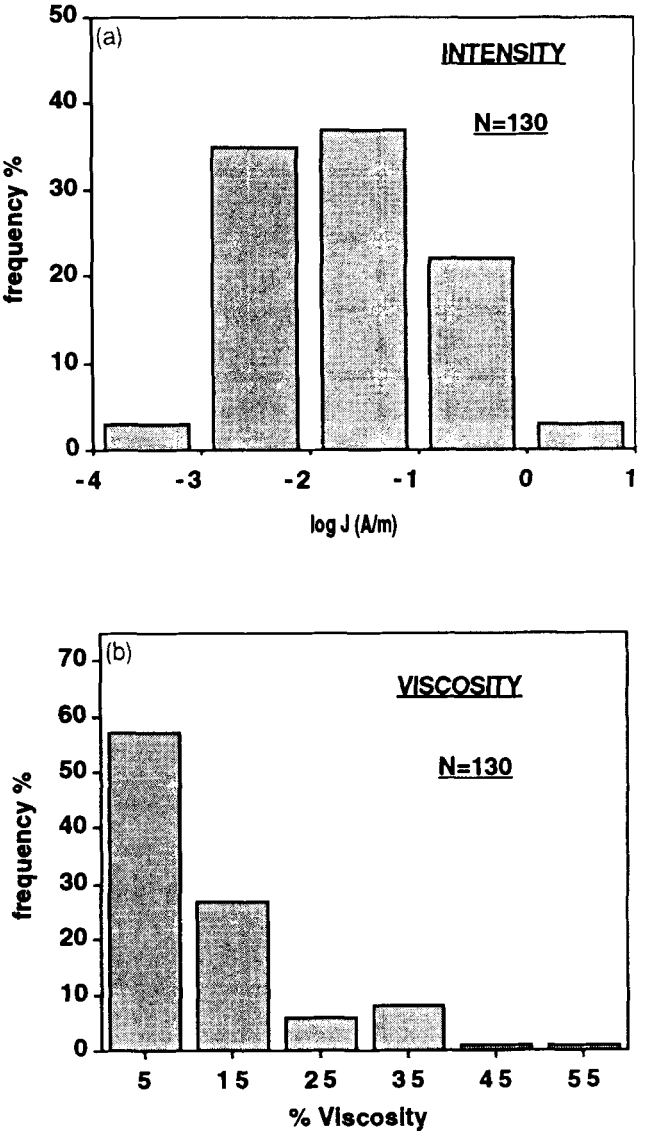


Figure 2. NRM results: (a) intensity, (b) viscosity (N = number of samples).

show three groups, the first being east with intermediate downward inclination with a trend to the dipolar present-day Earth magnetic field (EMF) at Sidi-Saïd-Maâchou ($D = 0°$, $I = +52°$), and the two others being north-west with shallow and intermediate upward inclination.

After tilt correction these directions are well grouped with an intermediate downward inclination for the east group and an intermediate upward inclination for the west group. As a conclusion, it can be anticipated that remanent magnetization has a composite nature.

One to three components of magnetization were revealed by each sample (Fig. 4), upon thermal or AF treatments. The first component was found in all samples and disappears usually by 150–250 °C or 10–30 mT. The direction for this component (A) in geographic coordinates is similar to the present-day EMF direction, indicating that this is most likely viscous magnetization of recent origin. Following removal of this low-temperature component, the results of samples

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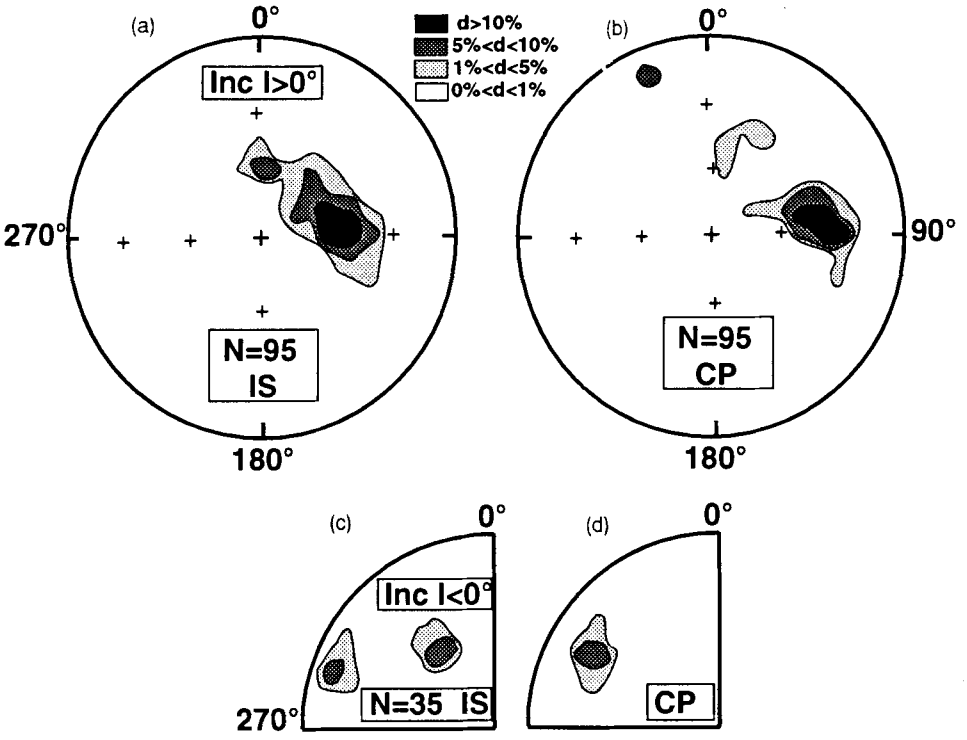


Figure 3. NRM results: direction density contour. Density of observations per unit area is represented by *d*.

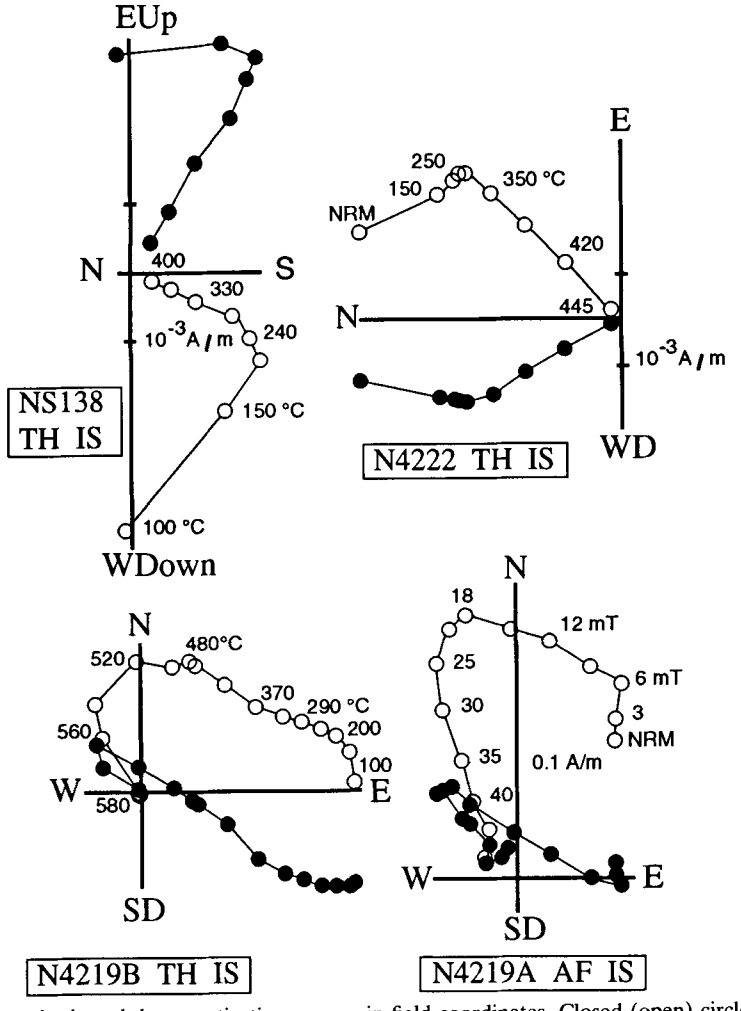


Figure 4. Orthogonal projections of selected demagnetization curves, in field coordinates. Closed (open) circles represent the projection onto the horizontal (vertical) plane. NRM intensities are given in amperes per metre, temperature in degrees Celsius and AF peak value in millitesla.

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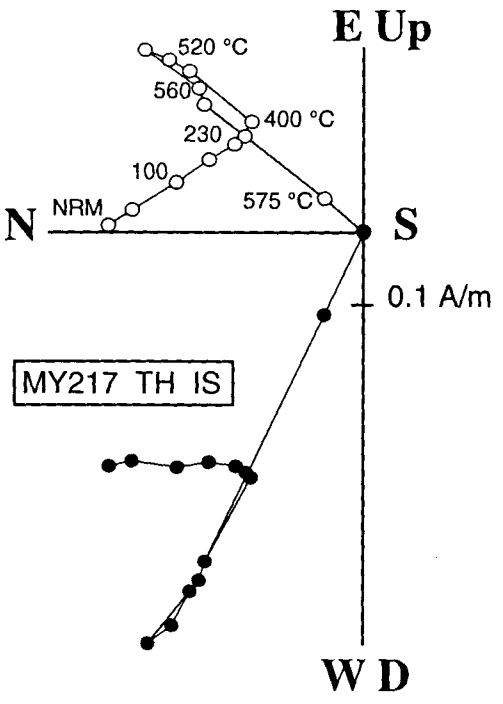


Figure 5. A demagnetization curve of a sample from site MY (same conventions as Fig. 4).

depicted in Fig. 4 show one or two other components of magnetization: a south-east direction with shallow to moderate downward inclination (B), which decays linearly to the origin by 480 °C (rarely 580 °C—sites NU and N2), or a north-west direction with shallow to intermediate usually upward inclination (C) by 580 °C (rarely by 500 °C—site N3 for example). When the two components B and C are observed in the same sample, B is always removed before C. In conclusion, there are three categories of samples:

(1) samples which carry the components of magnetization A + B (Fig. 4a) (50 per cent of samples);
(2) A + C (Fig. 4b) (30 per cent of samples);
(3) A + B + C (Figs 4c and 4d) (20 per cent of samples).

On top of these general features, some particularities were observed: (1) the sites NY and NZ display a large dispersion in directions, and thus were not used; (2) components (similar to the C component) with opposite polarity appear at the site MY (Fig. 5) after AF or thermal demagnetization.

MAGNETIC MINERALOGY

The methods used to identify the magnetic minerals in these rocks were microscopic investigations, IRM and thermomagnetic experiments.

Curie balance

Strong field thermomagnetic measurements were performed on crushed rock samples (100–150 mg) in air (A) or in vacuum (V), using an automatic recording Curie balance. If the paramagnetism is not important, the *Js* versus temperature (*Js–T*) curves are rarely reversible and show one or two Curie temperatures. The curves of most samples are similar to Fig. 6(b), with a single Curie point around 580 °C indicating the presence of a composition near magnetite with low titanium content. Fig. 6(c) shows a second Curie temperature in the range of 350–400 °C indicative of titanomagnetite. The *Js–T* in Fig. 6(d) is symptomatic of the presence of some maghemite (titanomaghemite), in addition to magnetite, and is characterized by the irreversible decrease of *Js* between 350 and 480 °C due to transformation of maghemite into haematite. Chemical alteration resulting from heating above 400 °C was observed in thermomagnetic experiments made on such samples in air (e.g. Fig. 6e).

In conclusion Curie temperatures in the ranges of 350–400 °C and 550–580 °C, indicative of maghemite (titanomaghemite), titanomagnetite and magnetite, must be noted. The presence of haematite in some samples is possible but is dominated by magnetite.

IRM

IRM acquisition experiments (Figs 7a and c) were performed on a representative number of samples which were demagnetized by AF. The IRM saturated samples were subsequently thermally demagnetized (Fig. 7b). The saturation and unblocking curves show that one to three phases of magnetic carriers are present in the samples:

(1) a high coercivity (>1.2 T) and high unblocking temperature (650 °C) phase corresponding to haematite;
(2) a high coercivity and very low unblocking temperature (150 °C) phase (goethite);
(3) low coercivity (<350 mT) and unblocking temperatures around 450 °C and 560 °C phases (titanomagnetites and magnetite);
(4) an unblocking temperature around 350 °C corresponding to maghemite.

Most of the curves reveal that titanomagnetites and magnetite are the predominant magnetic carriers.

Microscopic observations

Microscopic observations confirm this conclusion and indicate that pyrite, which sometimes seems a secondary mineral, is abundant in some samples and absent in others. It appears that there is a relationship between the presence of this mineral and the B magnetization as pyrite is absent in samples which contain only the C magnetization.

After these different rock magnetism experiments, including unblocking temperatures, it can therefore be concluded that the B component of magnetization is carried essentially by titanomagnetites and the C component by magnetite.

DISCUSSION OF THE RESULTS

Three groups of magnetization have been identified: low (A), intermediate (B) and high temperature (C) components. Tables 1, 2 and 3 list the results.

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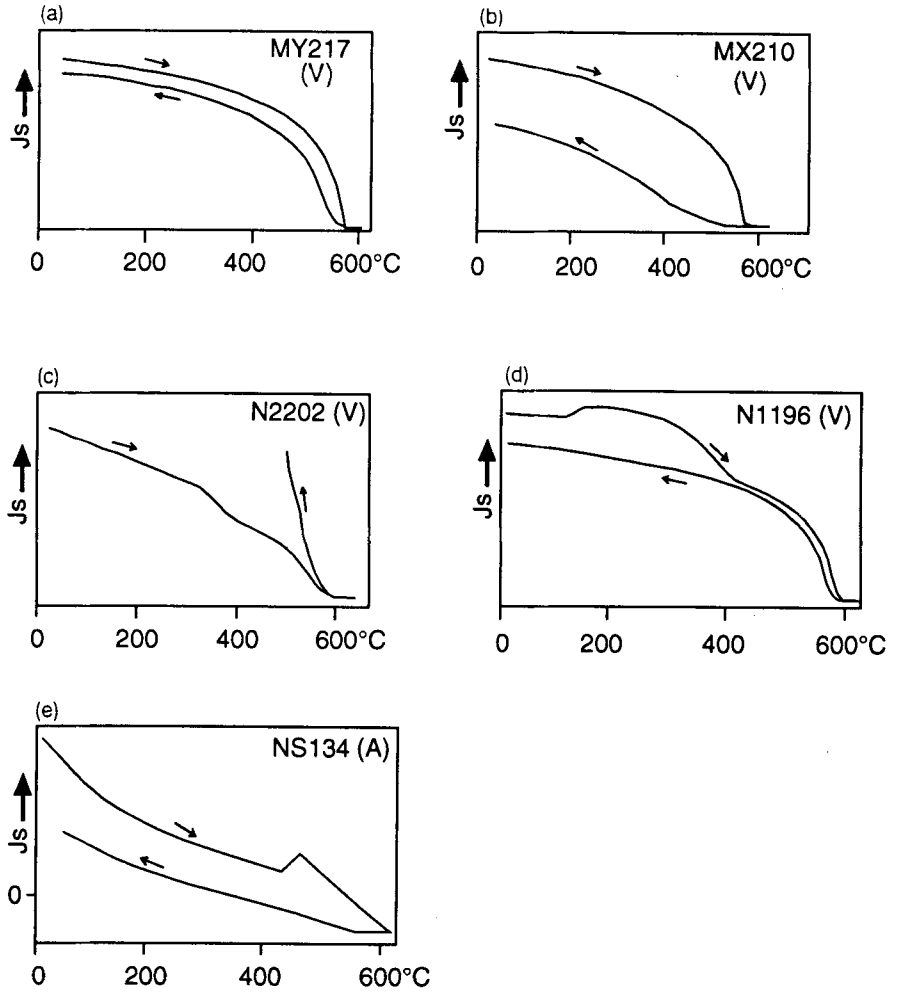


Figure 6. Thermomagnetic experiments for a representative set of samples. (V = Vacuum, A = air.)

Fold test

The classical fold test has been applied to the results to check whether the magnetizations were acquired before or after the Variscan folding. The site-mean directions before and after tilt correction are plotted in Figs 8 and 9.

Component A

The site-mean directions are well grouped (Fig. 8) before tilt correction with an average direction (declination/inclination = 8°/52°) similar to the dipolar present-day EMF at Sidi-Saïd-Maâchou ($D/I = 0°/52°$). This result confirms the recent origin of this magnetization.

The characteristic components B and C

The equal-area plots (Fig. 9) illustrate the effects of tilt corrections on site-mean directions for the characteristic components B and C.

Before tilt correction (Fig. 9a) we can distinguish for each magnetization B or C, the directions (with a shallow inclination) obtained from the M group sites (western dip) and those obtained from the N group sites (eastern dip).

The directions of the C component are very well grouped after tilt correction (Fig. 9b). Considering the 11 sites where C has been identified, the mean direction is $D = 300°$, $I = -31°$, $\alpha_{95} = 6°$. It presents a 99 per cent statistically significant positive fold test (the precision parameter k increase from 10 to 70) and corresponds to a palaeomagnetic pole at 14°N, 231°E ($\delta p = 4°$, $\delta m = 7°$).

The application of tilt correction to the B component does not display any significant variation of the k parameter. The B magnetization is a remagnetization which has partially overprinted the C magnetization. But in the case where B is observed alone, it could be a complete overprint or an opposite polarity of C. Thus the coexistence of the two cases could be at the origin of the fold test ambiguity. However, the examination of the two cases (B alone or with C) separately does not provide any conclusive evidence about the fold test.

It can be concluded that (1) the high-temperature magnetization (C) predates the deformation; (2) the dispersion especially in inclination before tilt correction and the relative more Fisherian distribution resulting from application of this tilt-correction lead us to suspect that the intermediate temperature magnetization (B) was also acquired prior to deformation, with a mean direction $D/I = 121°/+20°$ ($k = 33$; $\alpha_{95} = 6°$; $N = 17$) and a palaeopole at 18°S, 55°E ($\delta p = 3°$, $\delta m = 6°$).

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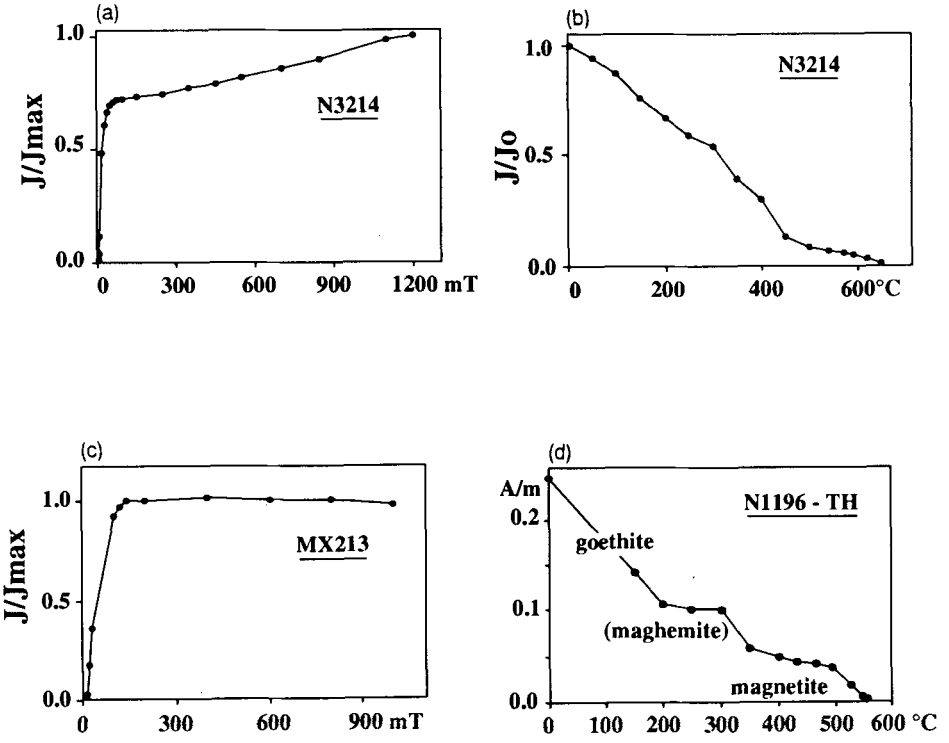


Figure 7. IRM curves (a, c) and thermal demagnetization of the IRM (b) and the NRM (d). J: intensity of magnetization, J_{max}: maximal intensity, J_0: initial intensity; A m^{-1}: amperes per metre.

Table 1. Low-temperature component. Bedding (S for strike and D for dip); Th is thermal treatment; AF is alternating fields; D is declination; I is inclination; IS is *in situ*; TC is after tilt correction; n is the number of samples (of sites in the mean); Fisher statistics (k for dispersion parameter, α_{95} for radius of circle of confidence at 95 per cent probability level).

Site	Bedding S/D	Cleaning range Th,°C Af,mT	IS D/I	TC D/I	n	k	α_{95}
Group N							
NQ	332/7NE	20-250 0-10	09/49	15/45	5	68	09
NR	000/17E	20-200 0-15	26/53	42/43	7	20	14
NS	005/13E	20-200 0-25	03/43	15/42	7	46	09
NTd	350/14E	20-200 0-25	12/46	24/39	6	20	15
NU	345/12E	20-200 0-07	11/42	19/36	7	14	16
NV	000/10E	20-200 0-10	11/49	22/46	5	57	10
NW	010/11E	20-200 0-07	13/49	24/45	7	59	08
NX	000/11E	20-250 0-15	05/57	21/54	7	23	13
N1	347/13E	20-200 0-15	00/36	08/32	5	06	35
N2	329/25NE	20-200 0-06	06/48	21/30	6	28	13
N4	006/40E	20-200 0-06	39/54	63/25	9	11	16
Group M							
MQ	187/23W	20-230 0-14	354/51	332/41	7	54	08
MS	174/22W	20-150 0-10	000/52	334/49	6	56	09
MT	232/29N	20-200 0-30	352/61	339/34	5	34	13
MU	153/29SW	20-160 0-06	004/45	331/52	5	11	24
MV	153/29SW	20-200 0-05	358/55	349/49	7	101	06
MW	145/56SW	20-150 0-20	355/49	287/44	7	12	28
MX	153/62SW	20-150 0-06	016/61	273/45	3	48	08
MY	177/19W	20-300 0-06	020/58	348/60	5	09	27
Mean							
Group N			012/48		11	76	05
				025/41	11	30	08
Group M			002/54		08	98	06
				324/50	08	16	14
M+N			008/51		19	71	09
				003/49	19	04	12

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Table 2. High-temperature component (C). Same conventions as Table 1.

Site	Bedding S/D	Cleaning range Th,°C AF,mT	IS D/I	TC D/I	n	k	α_{95}
Group N							
NQ	332/7NE	500-570 30-90	309/-35	305/-32	2	.	.
NTd	350/14E	450-550	304/-46	296/-35	4	.	.
NTe	350/14E	050-550 00-70	317/-40	309/-31	4	100	9
N1	347/13E	450-550 20-70	313/-44	304/-36	7	42	9
N3	356/42E	020-450 00-40	321/-51	300/-19	7	582	3
N4	006/40E	200-570 20-60	327/-47	310/-17	4	137	8
Group M							
MR	180/25W	020-585 00-95	291/-08	294/-31	9	396	3
MS	174/22W	480-585 43-95	295/-13	300/-31	7	376	3
MU	153/29SW	160-580 06-90	290/-12	299/-13	9	153	4
MX	153/62SW	150-520 20-55	278/+23	280/-29	5	76	9
MY	177/19W	300-560 06-27	294/-19	299/-36	8	281	3
Mean							
Group N			315/-44		6	96	7
				304/-29	6	70	8
Group M			290/-06		5	21	17
				294/-32	5	112	7
M+N			302/-28		11	10	15
				300/-31	11	70	6

Table 3. Intermediate-temperature component (B). Same conventions as Table 1.

Site	Bedding S/D	Cleaning range Th,°C AF,mT	IS D/I	TC D/I	n	k	α_{95}
Group N							
NQ	332/7NE	250-450 10-30	112/29	109/24	5	23	15
NR	000/17E	200-400 15-40	121/32	117/17	7	43	9
NS	005/13E	200-400 25-80	120/15	119/03	7	343	3
NTd	350/14E	200-450 20-50	119/25	116/14	6	381	3
NU	342/12E	200-580 07-35	128/25	124/18	7	105	6
NV	000/10E	200-450 07-35	112/28	110/19	5	59	10
NW	010/11E	300-450 07-45	117/39	114/29	7	92	6
NX	000/11E	250-420 10-30	110/38	108/28	7	52	8
N1	347/13E	200-450 10-20	117/13	116/03	5	62	10
N2	329/25NE	200-550 06-40	128/27	121/16	4	37	15
N4	006/40E	200-520 06-20	129/28	125/-7	5	78	6
Group M							
MQ	187/23W	230-430 14-75	121/12	125/33	5	133	7
MS	174/22W	150-480 10-43	130/13	136/27	7	376	3
MT	232/29N	200-420 30-80	120/-2	118/25	4	118	9
MW	145/56SW	150-490 20-80	121/4	131/20	6	132	6
MX	153/62SW	150-520 06-60	109/5	134/41	3	228	8
MY	177/19W	530-580 15-70	125/11	129/26	11	36	8
Mean							
Group N			120/27		11	68	6
				117/15	11	43	7
Group M			121/07		6	81	8
				129/29	6	76	8
M+N			120/20		17	36	6
				121/20	17	33	6

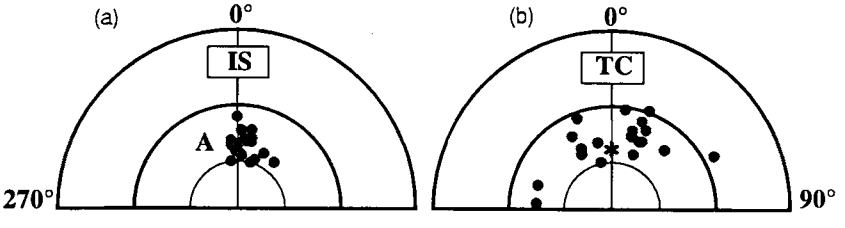


Figure 8. Stereographic projections of the site-mean directions of the low-temperature component (A): (a) before and (b) after tilt correction. Closed (open) circles represent projections onto the upper (lower) hemisphere. The asterisk represents the present-day Earth magnetic field.

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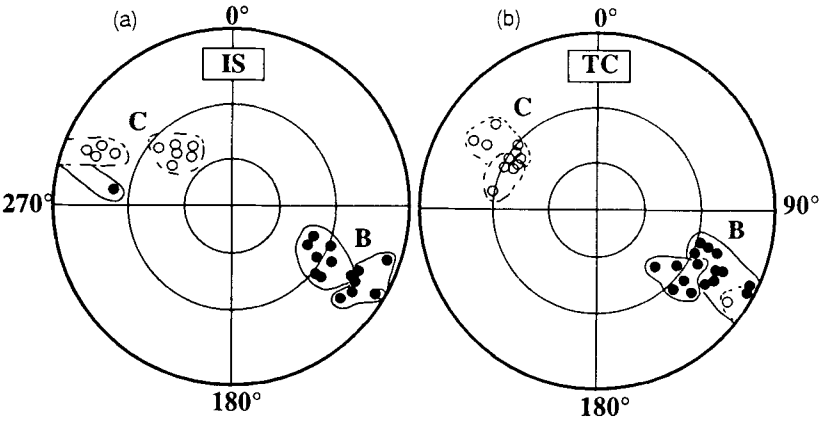


Figure 9. High (C) and intermediate (B) temperature components: (a) before and (b) after tilt correction (same conventions as Fig. 8).

Baked contact test

As noted above, a dyke (site NT) and the adjacent sediment were sampled to assess the age of magnetization in the dyke with baked contact test. The demagnetization results (Fig. 10) show that the dyke samples contain the two magnetizations (intermediate B and high temperature C). This behaviour contrasts with the univectorial magnetization similar (1) to C in the samples collected at the true margin of the dyke (samples 147 to 153); (2) to B magnetization in sample 154 collected 1 m from the dyke margin.

These results lead to the following interpretation: during its cooling the dyke and the baked sediment acquired the C magnetization. The univectorial magnetization in samples 147 to 153 is clearly consistent with a thermal origin, since the temperature near the dyke at the time of intrusion probably exceeded 580 °C, and thus was able to completely reset any magnetization carried by magnetite in these sedimentary samples.

Subsequently, a resetting of the characteristic magnetization during a thermal event (hydrothermal event) was partial in the dyke and complete in the unheated surrounding sediment host rock (sample 154). Thus the B magnetization was acquired. Moreover, as noted above, this magnetization is always accompanied by sulphides (pyrite).

Based on this evidence, the high-temperature C magnetization acquired before Variscan folding is therefore considered to be primary and of Middle Cambrian age (Acadian). The B magnetization, which is nearly antiparallel to C, appears to have been acquired a short time after.

Recent results of mineralogical, chemical and isotropic (K–Ar) studies on the Coastal-Meseta Cambrian greywackes (Rais 1992) can contribute to this discussion. They show: (1) a Variscan deformation which increases from NNW to SSE, and (2) three metamorphic events at 447 ± 12 Ma, 301 ± 8 Ma and 207 ± 6 Ma. In the Sidi-Saïd-Maâchou area, only the older event (447 ± 12 Ma) is found. Based on the crystallinity of illite, this event is linked to an anchizonal metamorphism ($T < 200$ °C). The Variscan event (301 ± 8 Ma), which is well developed in the SSE as mesozonal metamorphism, has a very low intensity (diagenetical metamorphism) in this area. This explains why the large Permo-Carboniferous (≈300 Ma) overprint, which is characterized in many regions of Morocco by a direction of magnetization about $D/I \approx 140°/0°$ (Najid 1986; Salmon *et al.* 1988; Khattach *et al.* 1989; Khattach & Najid 1994), is absent in the Sidi-Saïd-Maâchou volcanism. The B component could not be linked to the event at 447 Ma because of the low intensity of this event. Moreover, the palaeopole calculated from this B component is quite different from the African poles dated at 447 Ma. In conclusion, these new data confirm the interpretation about the age of the B and C components.

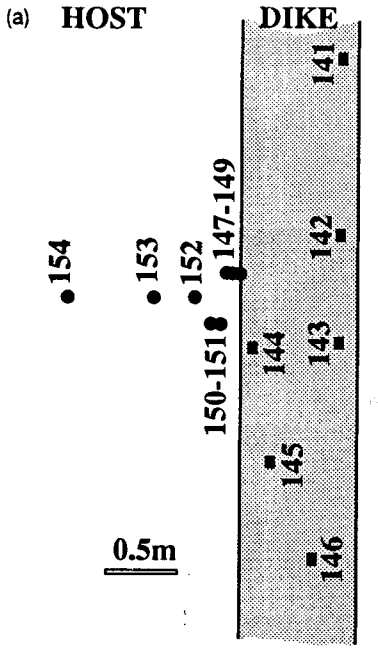


Figure 10. Baked contact test. The dyke and the locations of samples are shown schematically in (a); orthogonal projections of demagnetization curves for samples of the dyke (b), and of the host rock collected closest to the dyke margin (c) and farther from it (d).

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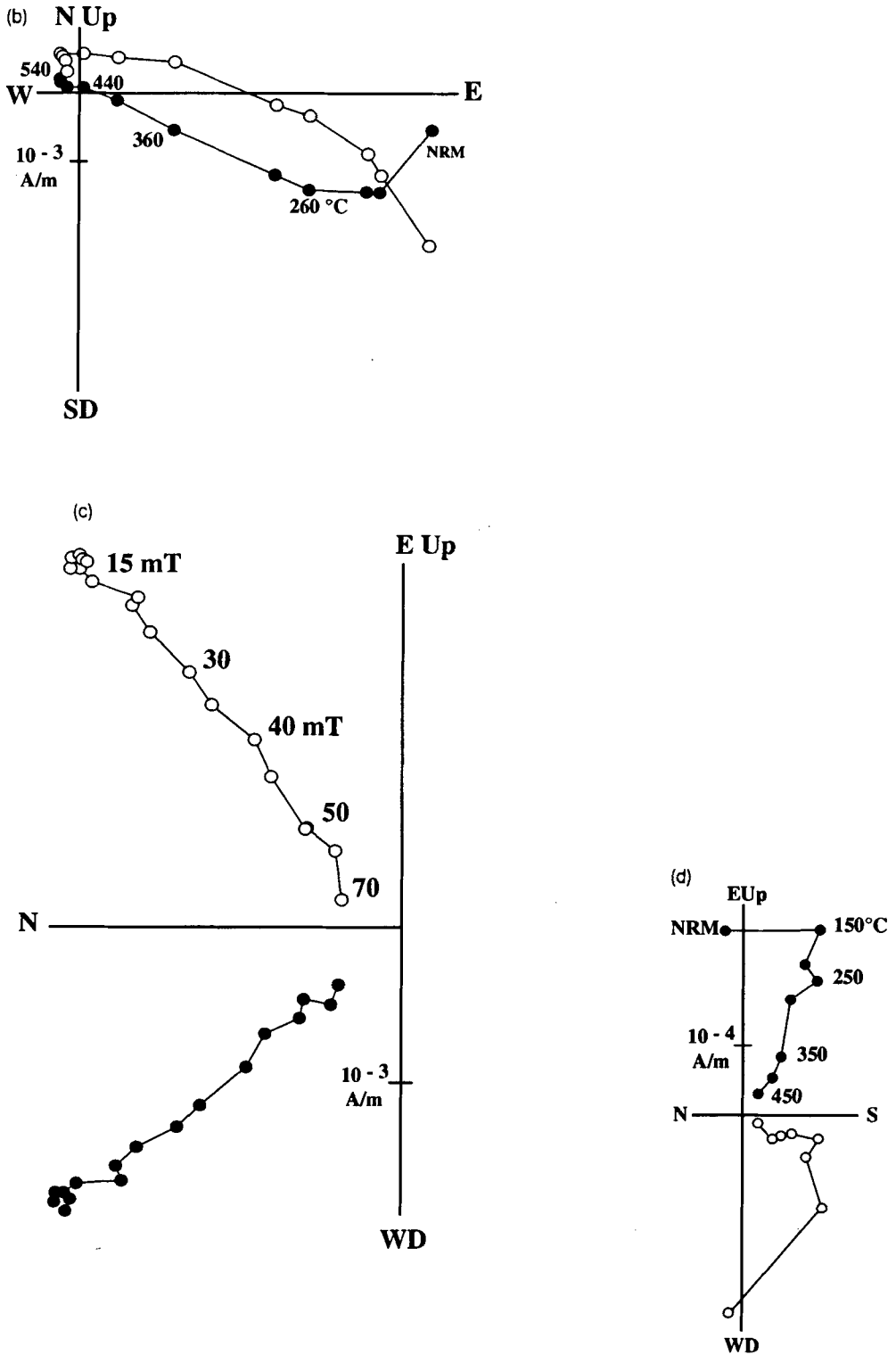


Figure 10. (*Continued.*)

CONCLUSIONS AND GEODYNAMIC IMPLICATIONS

Directions of magnetization observed within rock units from a given continental block are conventionally summarized by an apparent polar wander path (APWP) relative to the block. This path provides the basis from which changes in palaeolatitude and rotations of this block can be determined.

The Carboniferous and younger segments of the African APWP are known with reasonable accuracy (Irving & Irving 1982; Aïfa, Feinberg & Pozzi 1990; Besse & Courtillot 1991). In contrast, the lack of reliable palaeomagnetic poles for cratonic Gondwanaland for older periods has resulted in

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various and diverging models (see Van der Voo 1993 and the references listed therein) which strongly depend on the poles selected by each author. In a previous paper (Khattach *et al.* 1989), comparison of the Sidi-Saïd-Maâchou Cambrian pole with other African poles of about the same age led to the tentative conclusion that the Moroccan Coastal Meseta was possibly not linked with stable Africa during the Early Palaeozoic. This interpretation was also adopted by Feinberg *et al.* (1990) and therefore listed by Van der Voo (1993, Table A4). It is now considered that this possibility is not well argued since the comparison was chiefly made with palaeopoles which had not been clearly defined.

(1) The Hook Intrusives pole (14°N, 336°E, $k = 13$, $\alpha_{95} = 34$; Brock 1967) has a rather low quality factor ($Q = 3$; Van der Voo 1993, Table A4).

(2) For the Bou Azzer pole (47°N, 42°E, $k = 182$, $\alpha_{95} = 7$; Daly & Pozzi 1977), the authors did not suspect the Permian overprint which has been found almost everywhere in all types of rocks from the studied area (Hailwood & Tarling 1973; Najid 1986; Salmon *et al.* 1988; Khattach *et al.* 1989).

(3) The Nama Group pole from Namibia (pole N3; Kröner *et al.* 1980) corresponds to a remagnetization whose age is poorly constrained (Early Palaeozoic, 580–480 Ma). This pole could possibly be younger than the Sidi-Saïd-Maâchou pole, as those from the Oujeft Plateaux in Mauritania (pole PO; Morris & Carmichael 1978) and from Hassi Messaud in Algeria (pole HM; Bucur 1971) grossly correspond to the Cambrian–Ordovician transition.

Moreover, geological studies in the Moroccan Hercynides have shown that the southern and northern parts of Morocco were connected throughout the whole Palaeozoic, including the Cambrian period (Piqué *et al.* 1991 and references listed therein). It is therefore considered that the studied area of the Coastal Meseta was part of stable Africa (Table 4; Fig. 11) and the results of the present study have been combined with previous data (poles GW, SR, PC and AIR) already selected by Schmidt *et al.* (1990) on the basis of a modified version of Briden & Duff (1981) criteria and the earliest Cambrian pole AO defined from the Moroccan Anti-Atlas (Khattach 1989; Khattach *et al.* 1989). The

Table 4. Selected Lower Palaeozoic palaeomagnetic poles for Africa. Plat., Plong. are latitude and longitude of the palaeopole; α_{95} is the half-angle of the cone of 95 per cent confidence around the pole; δp, δm are the semi-axes of the elliptical error around the pole at a probability of 95 per cent, dp is the colatitude direction and dm perpendicular to it; Q is the quality classification (Schmidt *et al.* 1990). Age mnemonics: $\in$ is Cambrian, O is Ordovician, S is Silurian, l is lower, m is middle.

Rock formation	Mnemonic	Age (Ma)	Plat. (°N)	Plong. (°E)	α_{95}(°) δp δm	Q	Reference
Aïr ring complex	AIR	435	-43	9	6	A	Hargraves et al. (1987)
Packhuis/Cedarberg	PC	O-S	-25	163	18	B	Bachtadse et al. (1987)
Salala ring complex	SR	O	-40	150	9	B	Bachtadse & Briden (1989)
Graafwater Fm.	GW	Ol	28	14	9	B	Bachtadse et al. (1987)
Spilites Bou Regreg	SB	Om	-13	9	21	B	Khattach et al. (1989)
Ourzazate andesites	AO	535	-13	98	5 8	B	Khattach et al. (1989)
S.S.Maâchou volcanism	SSM	$\in$m	14	231	4 7	A	this paper

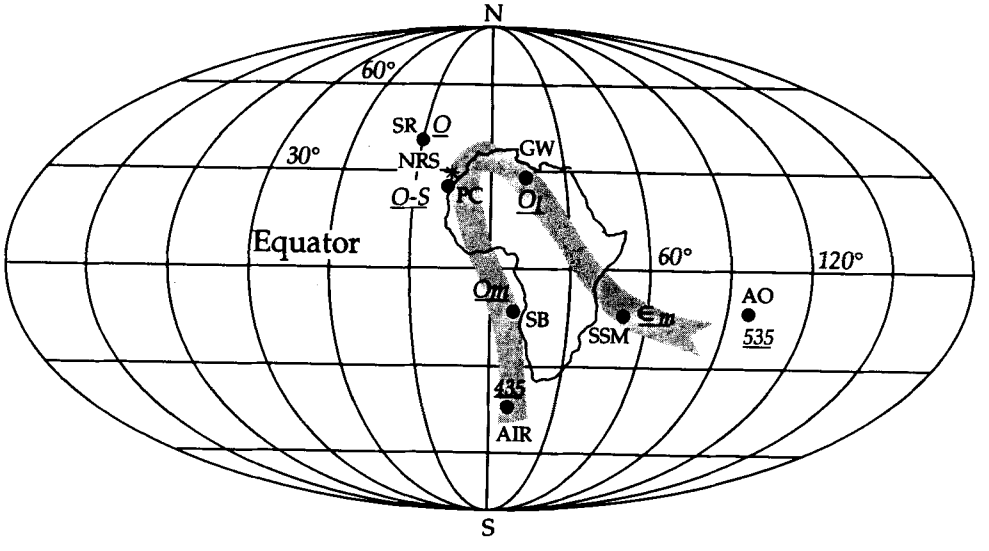


Figure 11. Proposed apparent polar wander path for the Lower Palaeozoic of Africa. Pole identifications correspond to Table 4 (pole NRS: The Ntonya Ring Structure, Briden 1968).

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Ordovician pole SB obtained from Bou Regreg spilites (Khattach 1989; Khattach *et al.* 1989) is also indicated in Fig. 11 and Table 4 but is not very well defined ($\alpha_{95} = 21°$).

The most notable feature of this APWP (Fig. 11) is that it confirms the relative movement of Africa during the Early Palaeozoic of about 15 cm yr^{-1}, as already proposed in previous models. This APWP requires that the pole moved from the south-eastern margin of Africa in the Middle Cambrian, to North Africa by Ordovician time and back to south-western Africa by Late Ordovician and Silurian times. The rapid polar movement is consistent with the migration of glacial centres across Gondwanaland suggested by Caputo & Crowell (1985) from geological information.

It must be mentioned that the Ntonya Ring Structure from Malawi (Briden 1968) has provided results indicating a pole position in north-western Africa (pole NRS: 28°N, 15°W; see Fig. 11). A recent re-examination of the Ntonya Complex (Briden, McClelland & Rex 1993) led to the conclusion that the magnetization was acquired at 522 ± 13 Ma, which is apparently contemporaneous with the age of the Sidi-Saïd-Maâchou volcanics. There is, at present, no solution to this contradiction but the problem would possibly be cleared up by more precise calibration of the Cambrian time-scale. However, it can be added that the occurrence of tillites within the purple Ahnet Series of western Hoggar (see Fabre 1983, p. 181) can hardly be regarded as corroborating a high latitudinal position of this region during Middle Cambrian times (Briden *et al.* 1993) as the precise age of the Ahnet Series is not clearly established (see Legrand 1985, p. 23).

During Middle Cambrian times, the western part of the Moroccan Meseta was a subsiding area characterized by thick mainly terrigenous deposits (Destombes, Hollard & Willefert 1985). Field data and geophysical investigations (Bernardin *et al.* 1988) indicate that this area was a large graben (400 km long and 80 km wide) extending at least from the western part of the Haut Atlas to the coastal Meseta of Casablanca. This structure was controlled by major faults oriented N30°E and limited eastwards by the so-called West Moroccan Shear Zone (Piqué *et al.* 1991). The whole area was characterized by similar thick terrigenous deposits of shallow-water facies. Volcanic activity within this rather complex area is known in the Haut Atlas, at Sidi-Saïd-Maâchou and in the Oued Rhebar area. The geodynamical significance of this structure is not completely elucidated and might correspond either to a passive margin or to a rift (Cornée, Destombes & Willefert 1987).

Similar distensive structures are also known in Nova Scotia (Avalon Zone) where Lower and Middle Cambrian volcanics occur along the south-eastern side of the Appalachians and might correspond either to a volcanic arc or to a rift (Murphy *et al.* 1985). On the other hand, palaeomagnetic studies of the Middle Cambrian Bourinot Group indicate that Nova Scotia was not part of cratonic North America at that time (Johnson & Van der Voo 1985). This is fully consistent with palaeontological data which show a clear distinction between the Cambrian benthic faunas of cratonic North America (Laurentia) and those of Avalonia which are of 'Atlantic' type and have much in common with those of Baltica and western Gondwana (see McKerrow, Scotese & Brasier 1992; Courjault-Radé, Debrenne & Gandin 1992 and listed references). In the palaeogeographic reconstructions proposed for the Cambrian, Avalonia is therefore placed along the western margin of Gondwana. The Middle Cambrian palaeolatitude of the Moroccan Coastal Meseta (≈15°S, this paper) significantly differs from that of Nova Scotia (49°S, Johnson & Van der Voo 1985), which also corresponds to distinct characteristics of Cambrian successions in Morocco (the presence of archeocyathid limestones) and in Avalonia (the dominance of clastic sediments). All this suggests that Avalonia was situated further south along the western Gondwana margin. The distensive structures and volcanic activity, which occur both in the Moroccan Coastal Meseta and Nova Scotia, would correspond to rifting processes operating along the Gondwana margin and announcing the separation of Avalonia from Gondwana.

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